

Table of Contents

Bitwise Operations

1. 1146 — Counting Bits
2. 1655 — Maximum Xor Subarray
3. 3191 — Maximum Xor Subset
4. 3211 — Number of Subset Xors
5. 3192 — *K Subset Xors (no solution found)*
6. 3233 — *All Subarray Xors (no solution found)*
7. 2419 — Xor Pyramid Peak
8. 3194 — Xor Pyramid Diagonal
9. 3195 — *Xor Pyramid Row (no solution found)*
0. 1654 — *SOS Bit Problem (no solution found)*
1. 3141 — *And Subset Count (no solution found)*

CSES Problem Set

Counting Bits

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to count the number of one bits in the binary representations of integers between 1 and n .

Input

The only input line has an integer n .

Output

Print the number of one bits in the binary representations of integers between 1 and n .

Constraints

- $1 \leq n \leq 10^{15}$

Example

Input:
7

Output:
12

Explanation: The binary representations of $1 \dots 7$ are 1, 10, 11, 100, 101, 110, and 111, so there are a total of 12 one bits.

CSES Problem Set**Maximum Xor Subarray**TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

Given an array of n integers, your task is to find the maximum xor sum of a subarray.

Input

The first line has an integer n : the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the maximum xor sum in a subarray.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq x_i \leq 10^9$

Example

Input:

```
4
5 1 5 9
```

Output:

```
13
```

CSES Problem Set

Maximum Xor Subset

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an array of n integers, your task is to find the maximum xor sum of a subset.

Input

The first line has an integer n : the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the maximum xor sum of a subset.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq x_i \leq 10^9$

Example

Input:

4
1 6 12 6

Output:

13

CSES Problem Set

Number of Subset Xors

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given an array of n integers, your task is to find the number of different subset xors.

Input

The first line has an integer n : the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the number of different subset xors.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq x_i \leq 10^9$

Example

Input:

```
3
3 6 5
```

Output:

```
4
```

Explanation: The following values can be the xor of a subset:

- $0 = \text{xor of the empty set}$
- $3 = 3$
- $5 = 3 \oplus 6$
- $6 = 3 \oplus 5$

In this case, no other values can be the xor of a subset.

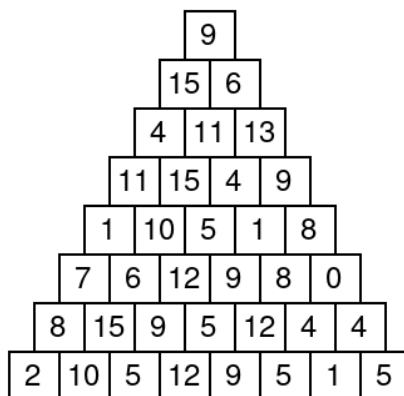
CSES Problem Set

Xor Pyramid Peak

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a xor pyramid where each number is the xor of lower-left and lower-right numbers. Here is an example pyramid:



Given the bottom row of the pyramid, your task is to find the topmost number.

Input

The first line has an integer n : the size of the pyramid.

The next line has n integers $a_1, a_2, \dots, a_n$: the bottom row of the pyramid.

Output

Print one integer: the topmost number.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a_i \leq 10^9$

Example

Input:

8
2 10 5 12 9 5 1 5

Output:

9

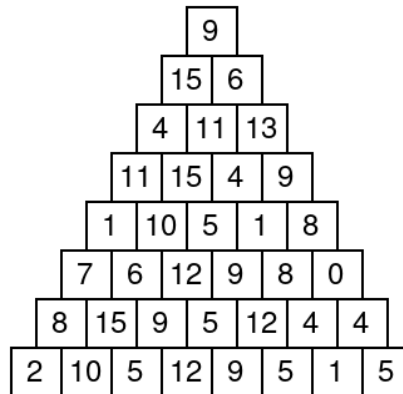
CSES Problem Set

Xor Pyramid Diagonal

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a xor pyramid where each number is the xor of lower-left and lower-right numbers. Here is an example pyramid:



Given the bottom row of the pyramid, your task is to find the leftmost number of each row.

Input

The first line has an integer n : the size of the pyramid.

The next line has n integers $a_1, a_2, \dots, a_n$: the bottom row of the pyramid.

Output

Print n integers: the leftmost numbers of the rows from bottom to top.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a_i \leq 10^9$

Example

Input:

```
8
2 10 5 12 9 5 1 5
```

Output:

```
2 8 7 1 11 4 15 9
```